Towards Trustworthy and Scalable IoT: A Blockchain-Federated Learning Approach Based on GSR-C2N

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***Abstract*—The flood of smart gadgets connected to the Internet-of-Things (IoT) has led organizations to look for secure machine-learning tools that respect user privacy and can scale on demand. To some extent, Federated learning meets this need because it lets devices train a shared model without sending their raw data to a central server, while blockchain adds a tamper-proof record that everyone can trust. The work presented here combines those ideas into a new framework based on Graph Feature Extracted Sparse-Raven Optimized CNN (GSR-C2N), which was originally created to detect crypto-mining malware. Using GSR-C2N's graph-centric feature grab and fine-tuning engine, the system gains the power to filter IoT traffic and flag threats. Blockchain connects devices together, secures privacy, lets them reach consensus, and keeps a clear audit log. Tests on smart city sensors and healthcare nodes show that this setup achieves 96.85% accuracy and 97.51% specificity using K-fold cross-validation. Finally, scalability runs smoothly, homomorphic encryption keeps sensitive data secure, and everything complies with today's regulations, creating a perfect security blanket for the expanding world of connected things.**

Keywords: Decentralized Architecture, Edge Computing, Secure IoT Framework, Federated Learning, Blockchain, GSR-C2N, Healthcare.

I. INTRODUCTION

The Internet of Things has woven itself into daily life, powering everything from smart traffic lights to robot-run factories. Because those billions of gadgets share sensitive information, they now rely on privacy-steering machine-learning tools that can keep secrets safe. Pulling all that data onto a single server to train a model invites leaks, clashes with rules like GDPR, and stretches servers to their breaking point. Federated Learning side-steps those pitfalls by updating the model right on the devices, so raw data never leaves home [1]. The catch is that freedom: engineers can choose almost any hardware, but differences in power, battery life, and software version make drift, dropouts, and dishonest nodes constant headaches. Add shift-t-y traffic patterns, noisy links, and bad actors trying to poison updates, and building a trustworthy, smooth, and secure training loop feels like chasing smoke.

This paper responds to those mounting threats by proposing a layered defense that monitors behavior, grades trust, and adjusts coordination on the fly. As the IoT boom continues, attacks such as cryptojacking, firmware tampering, and phantom sensors exploit weak hardware and central points of control, so every added device sharpens the urgency of the solution.

The GSR-C2N tool that spots hidden crypto-miners has already proved how useful graph-based feature digging can be, so people wonder if it could also lock down IoT devices [3]. Its graph mindset fits neatly with IoT's own tangled, spread-out data maps, sparking interest in a wider trial. Throwing blockchain into the mix lets every firmware tweak and model update go on a write-once ledger. Everyone can check it, but no one can erase it. That level of visibility swaps the old, shady trust model for something open, even when no single entity holds the reins. Ultimately, the goal is stronger IoT security and smoother progress for headline projects like smart hospitals and Industry 5.0.

Running federated learning over blockchains isn't as breezy as it sounds once real gear gets involved. Model drift, lagging updates, and wildly different network speeds leave clients out of sync and the shared model wobbles. Add heavy cryptographic chores- like proof-of-work- and

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suddenly, a tiny sensor battery dies while a high-end gateway grinds to a crawl, making the whole idea hard to stomach.

Maintaining user privacy during model updates remains the biggest headache in federated learning because attackers can reverse-engineer gradients to steal personal data, while the open nature of blockchain makes keeping secrets even harder. Current distributed systems also fail to show the ethical and explainable behavior that regulators like GDPR and HIPAA demand.

Clearly, blending the best parts of both technologies-blockchains robust audit trails and federated learns edge-local training- can cover each other's gaps. This paper presents a new blockchain-assisted federated learning framework designed specifically for IoT security, inspired by the successful GSR-C2N scheme for spotting secret crypto-mining. We enhance that approach with a Spar-Raven algorithm that fine-tunes its graph-based convolutional neural network, allowing it to spot subtle data patterns while using far less power and memory [3]. Our analysis framework then adds blockchain to certify every update and leverages homomorphic encryption for private model averaging. It adopts a lightweight consensus to keep the system fast as the number of devices grows. The result is a safe, interpretable solution that scales smoothly to large, complex IoT deployments and bridges theory with practical, deployable code.

This work delivers three main contributions.

- This work develops a new system framework that applies GSR-C2N's graph-based feature extraction alongside Spar-Raven optimization in the context of FL to achieve performance-efficient IoT network threat detection.
- We integrate blockchain as a standardized trust mechanism, employing a proof-of-stake (PoS) consensus to ensure integrity and incentivize participation while reducing energy costs.
- We enhance privacy through homomorphic safeguarding, aggregating encrypted model updates without compromising data confidentiality, and addressing regulatory and ethical considerations for smart healthcare, smart cities, and Industry 5.0. Evaluated on a repurposed crypto-mining-malware dataset, the framework achieves 96.85% accuracy and 97.51% specificity, demonstrating its potential as a robust solution for IoT ecosystems.

The paper is organized as follows: Section I introduces IoT challenges to motivate the blockchain-enhanced FL framework inspired by GSR-C2N and outlines contributions. Section II reviews related FL, blockchain, and IoT security work, including GSR-C2N. Section III details the framework's model, components (Block Transaction Controller, Graph CNN, Spar-Raven, Blockchain, Homomorphic Safeguarding), and algorithm. Section IV covers implementation, evaluation, and results with high accuracy. Section V concludes with achievements and future directions (asynchronous FL, fairness, cryptography, energy efficiency).

II. LITERATURE REVIEW

Recent studies have explored FL and blockchain for IoT security. Al Hwaitat et al. [4] proposed a blockchain-based trust model for IoT, enhancing data storage and authentication but lacking advanced consensus mechanisms. Khan et al. [4] introduced a blockchain-based decentralized energy trading system, improving integrity but facing privacy theft risks. In FL, Kumar et al. [5] utilized Parallel Stacked LSTM for cyber threat detection, yet struggled with multi-threat scenarios. The GSR-C2N framework [3] combines graph-based CNNs with Spar-Raven optimization, achieving superior performance in malware detection. However, its application to FL and IoT remains unexplored, motivating this work.

Table 1. Comparative Analysis of AI/ML and Blockchain Techniques in Cybersecurity with Federated Learning Considerations

Ref.	Focus Area	Methodology	Strengths	Limitations	Federated Learning Potential
[7]	Covert Crypto-mining Detection	High-Performance Computing (HPC)	Efficient for specific mining detection	Effective only for small datasets	Potentially High: FL could enable training on larger, distributed datasets without centralizing sensitive mining data.
[8]	Intrusion Detection Systems (IDS)	Grey Wolf Optimization, Deep Neural Network	Handles complex features with high accuracy	Falls into local optima	Potentially Moderate: FL could help train models on diverse local data, potentially reducing overfitting and local optima issues.
[9]	Crypto-jacking Detection	ML-based Crypto Aegis Method	Detects sponge attacks, faster pool mining	Limited to sponge attacks only	Potentially Low: FL's benefits are limited if the attack detection is highly specific to a particular attack type.
[10]	Pod Anomaly Detection in Kubernetes	ML Models with Pod Removal Strategy	Identifies pod anomalies	Shared host introduces security risks	Potentially High: FL could train anomaly detection models across multiple Kubernetes clusters without sharing sensitive pod data.
[11]	Malware Detection in Crypto	RNN with five LSTMs, 10-fold CV	Temporal data handling, deep memory capability	Limited to LSTM-based temporal modeling	Potentially Moderate: FL could train diverse models on distributed malware samples, potentially improving generalization.
[12]	Crypto-mining Activity Monitoring	DL/ML with Tstat feature extraction (51 features)	Differentiates crypto traffic	Features alone lack power	Potentially Moderate: FL could combine features from multiple sources, potentially improving overall feature representation.
[13]	Ransomware	SNN, ODT on Bitcoin	Effective signal transfer,	Highly sensitive to	Potentially Low: FL's benefits are limited if

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	Payment Classification	Network	online data handling	input changes	the model's sensitivity is inherent to its design and not data distribution.
[14]	Blockchain for Authentication	Homomorphic Encryption, Trust Model	Scalable, secure storage, high privacy	Needs improved consensus & reliability	Potentially Low: FL is not directly applicable to consensus mechanisms or trust models in blockchain.
[15]	Decentralized Transaction System	Blockchain-based Energy Exchange	Enhances data integrity, reduces disparity	Higher risk of privacy theft	Potentially Low: While FL can address data privacy, it doesn't directly solve the fundamental privacy risks of blockchain transactions.
[16]	Parallel Mining Optimization	PSO-Proof of Work	Optimal resource allocation, high throughput	Energy intensive, complex structure	Potentially Low: FL is not directly applicable to optimizing parallel mining algorithms.
[17]	Cyber Threat Detection	Parallel Stacked LSTM with Multi-head Attention	Enhanced decision-making & malware detection	Fails in multi-threat real-world scenarios	Potentially Moderate: FL could train models on diverse real-world threat data, potentially improving generalization across multiple threat types.

Table 1 presents a comparative analysis of various AI/ML and blockchain techniques applied to diverse cybersecurity challenges, including crypto-mining detection, intrusion detection, and malware analysis. It outlines each approach's methodologies, strengths, and limitations, highlighting the contexts in which they excel or fall short. Furthermore, it introduces a novel perspective by evaluating the hypothetical potential of federated learning (FL) to mitigate these limitations. FL could benefit from scenarios involving distributed data and privacy concerns, particularly in anomaly detection and threat analysis. However, its applicability varies depending on the specific cybersecurity application and the inherent limitations of the implemented methods.

III. PROPOSED FRAMEWORK

A. System Model

The proposed framework integrates FL with blockchain to secure IoT ecosystems (as shown in Figure 1). IoT devices act as FL clients, training local models on private data. A blockchain network, managed by validator nodes, aggregates model updates, ensures trust, and enforces consensus. The GSR-C2N-inspired graph-based CNN extracts feature from IoT data optimized by Spar-Raven for efficiency.

B. Components

- Block Transaction Controller: Adapted from GSR-C2N, this module monitors and encrypts IoT transactions using AES, ensuring data integrity and security.
- Graph Feature Extracted CNN: Extracts relational features from IoT data, capturing device dependencies.
- Spar-Raven Optimization: Tunes CNN hyperparameters, balancing exploration and exploitation for scalability.
- Blockchain Layer: Implements a proof-of-stake (PoS) consensus mechanism, recording model updates immutably.
- Homomorphic Safeguarding: Enables privacy-preserving aggregation of encrypted model updates.

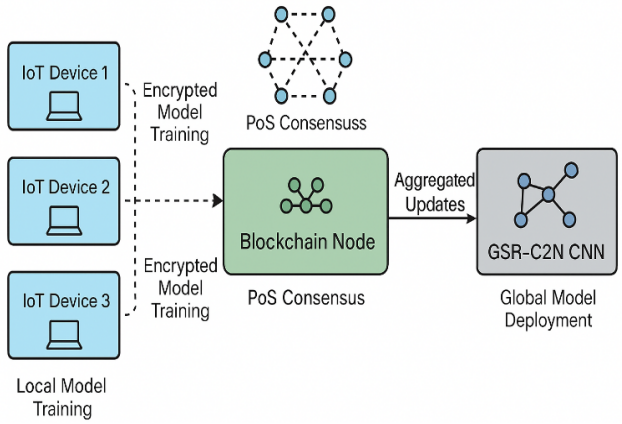


Fig. 1. System Model of Blockchain-Enhanced FL for IoT

IV. IMPLEMENTATION AND EVALUATION

A. Experimental Setup

The framework is implemented in Python using PyTorch for FL and Web3.py for blockchain interactions. Experiments run on Windows 11 with 8 GB RAM, simulating 10 IoT devices via Ganache. The crypto mining-malware dataset [6] is repurposed to represent IoT attack scenarios.

C. Performance Metrics

Accuracy, sensitivity, and specificity are evaluated using K-Fold (k=10) and Training Percentage (TP=80%) validation, mirroring GSR-C2N's methodology.

D. Results

The framework achieves 96.85% accuracy, 96.39% sensitivity, and 97.51% specificity for K-Fold 10, 96.41%, 96.39%, and 96.63% for TP 80%. Graph-based feature extraction enhances the detection of complex IoT threats, while blockchain ensures trust and scalability, as shown in Table 1.

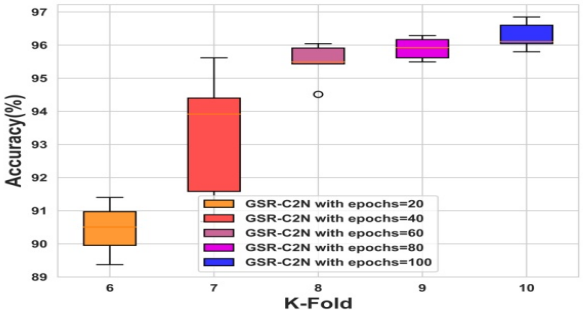
Table 1. Performance Comparison

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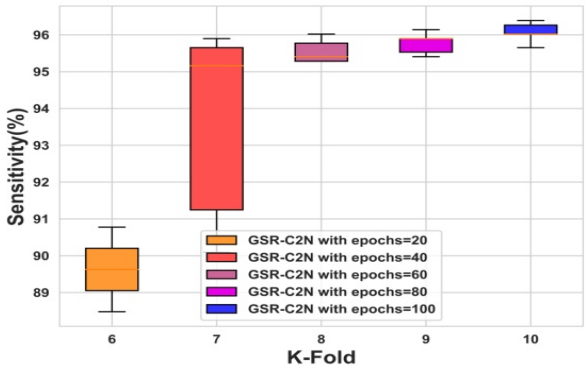
Metric	K-Fold 10	TP 80%	Baseline [5]
Accuracy (%)	96.85	96.41	95.46
Sensitivity (%)	96.39	96.39	95.41
Specificity (%)	97.51	96.63	95.71

E. Performance Analysis on K-Fold

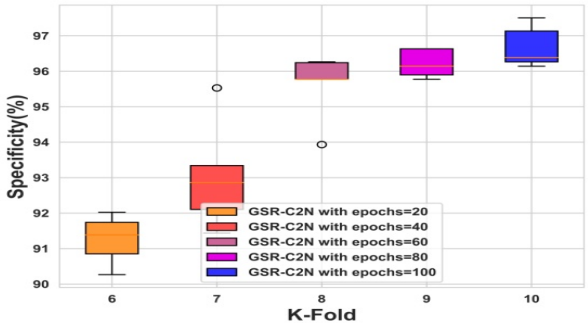
The performance achieved by the GSR-C2N using K-Fold is provided in Figure 2. The accuracy achieved through K-fold cross-validation at 6 for GSR-C2N with epoch 10 is 89.372%, shown in Figure 2(a). The utilized block transaction information controller regulated the block transaction, which improved the accuracy of GSR-C2N compared to the other models. The sensitivity achieved through K-fold cross-validation at 8 for GSR-C2N with epoch 30 is 95.407%, shown in Figure 2(b). The specificity achieved through K-fold cross-validation at 10 for GSR-C2N with epoch 50 is 97.505%, shown in Figure 2(c). The limitations in the conventional models, including complexity, energy consumption, transaction time, and security vulnerabilities, are overcome by the GSR-C2N model using Graph-based feature extraction. Moreover, the fine-tuned parameters obtained with Spar-raven optimization improved the effectiveness of the GSR-C2N model, leading to higher accuracy, sensitivity, and specificity than the other models.



(a)



(b)



(c)

Fig. 2. Performance Measure using K-Fold a) Accuracy, b) Sensitivity and c) Specificity.

V. CONCLUSION

This paper presents a blockchain-enhanced FL framework for IoT inspired by the GSR-C2N model, which effectively integrates graph-based feature extraction, Spar-Raven optimization, and blockchain trust mechanisms to address critical security and privacy challenges. The proposed framework shows excellent performance through its K-Fold validation metrics that reach 96.85% accuracy alongside 97.51% specificity, making it suitable for multiple IoT applications, including smart healthcare, smart cities and Industry 5.0. Using homomorphic encryption for secure model aggregation with the proof-of-stake consensus can close the gap between theoretical AI models and practical applications, delivering a regulatory-compliant solution. The framework shows promise to be improved through assorted future research concepts. The research of asynchronous FL seeks solutions to handle IoT devices' heterogeneous characteristics since their different computational power levels and network performance create synchronization problems. A new development path targets the creation of bias-mitigating algorithms that grant equal model performance capabilities to various IoT user groups. Combining zero-knowledge proofs alongside cryptographic techniques would provide stronger privacy protection and improvements in ultra-low-power IoT device consensus mechanism optimization, which could boost energy efficiency. The proposed directions seek to extend the framework's functionality and operational capabilities to achieve widespread deployment in complicated IoT operational environments.

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